Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

AI Security for Autonomous AI Agents

TGS Ref No: TGS-2025060473

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 5.0

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	16 June 2025	Initial release.	Dr Alfred Ang
2.0	17 August 2026	Retitled and rebuilt for autonomous-agent security; five activities added. TSC K/A text unchanged.	Dr Alfred Ang
2.1	18 August 2026	Added prompt-injection, PDPA, guardrail, approval, skill/plugin and visual coverage. TSC K/A text unchanged.	Dr Alfred Ang
3.0	20 August 2026	Evidence-grounded 207-slide rebuild with product boundaries, cases, controls, activities and source register. TSC K/A text unchanged.	Dr Alfred Ang
4.0	25 August 2026	Rebuilt as a 1-day course with three LO-aligned topics, eight no-code activities and a reflection-based practical assessment. TSC K/A text unchanged.	Dr Alfred Ang
4.1	25 August 2026	Refreshed trainer visuals and embedded the YouTube Online Video; scope, timing and TSC K/A text unchanged.	Dr Alfred Ang
4.2	25 August 2026	Added coding-harness and latest-AI-agents reference slides to the deck (Claude Code, Codex, DeepSeek Harness; OpenClaw, Hermes, Prime Agent, OpenWorker, QM). Schedule and TSC K/A text unchanged.	Dr Alfred Ang
4.3	25 August 2026	Beefed up Topic 3 with Singapore AI case studies (adoption vs cybersecurity, gen-AI brand damage, agentic testing, healthcare AgentSea), a brand-damage discussion/debrief, a 'Key Challenges of AI to Business' capstone and a visual multi-agent map. Schedule and TSC K/A text unchanged.	Dr Alfred Ang
4.4	25 August 2026	Activity 5 rebuilt around the AI Data Policy Generator website (learner-supplied training API key, 7-part IMDA/PDPA-aligned policy framework). Slides, LG and activity pack aligned; added a worked-example slide with concrete Scope, Roles, Rules and Review examples. Schedule timing and TSC K/A text unchanged.	Dr Alfred Ang
4.5	25 August 2026	Added Topic 1 slide 'Probabilistic Output — Why Traditional Security Struggles' contrasting deterministic software with probabilistic LLM output and the challenge to rule-based defences. Schedule timing and TSC K/A text unchanged.	Dr Alfred Ang
4.6	25 August 2026	Added Topic 1 slide 'When the Loop Goes Wrong — Misreading the Goal' after the concrete agentic-loop example: an open-ended 'remove the malware at all costs' goal escalates to wiping every file, motivating the Topic 3 guardrails. Schedule timing and TSC K/A text unchanged.	Dr Alfred Ang
4.7	25 August 2026	Added Topic 1 slide 'An Agent on WhatsApp Is Exposed to the Internet' after the TIA Support slide: a public messaging agent accepts messages from anyone, so prompt injection and data poisoning can extract confidential data or harm the server. Schedule timing and TSC K/A text unchanged.	Dr Alfred Ang
4.8	26 August 2026	Added verified case study 'Case — A Malicious Skill on ClawHub' (Snyk, Feb 2026) at the end of the Topic 1 Skills and Tools section: a fake Google skill whose SKILL.md instructions tricked users into installing malware. Schedule timing and TSC K/A text unchanged.	Dr Alfred Ang
4.9	26 August 2026	Activity 3 reworked: the agent now creates a 10-slide 'AI Security for AI Agents' deck (saved to the Downloads folder), then redesigns it from an uploaded image template — replacing the Envato-template animated-deck flow. Schedule timing and TSC K/A text unchanged.	Dr Alfred Ang
4.10	26 August 2026	Activity 2 reworked: learners now upload MOCK-MARKETING-DATA.xlsx to the Hermes + MiniMax agent and prompt it to	Dr Alfred Ang

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		analyse the data, generate charts and produce a .docx report with insights, analysis and recommendations. Schedule timing and TSC K/A text unchanged.	
4.11	26 August 2026	Added four Topic 1 slides on applications of AI agents in cyber security: threat detection (real-time detection, automated vulnerability analysis, malware classification), network and social-engineering security (intrusion detection, attack classification, phishing defence, log anomaly detection), incident response (patch generation, playbooks, root-cause analysis) and Security Operations Centres (threat hunting, predictive analytics, adaptive decision-making). Learner Guide notes renumbered. TSC K/A text unchanged.	Dr Alfred Ang
4.12	26 August 2026	Added Topic 1 slide 'Agent-to-Agent Messaging - A Security Blind Spot' after the multi-agent worked example: bad or malicious output cascades agent-to-agent, and a growing topology multiplies message paths - reducing visibility and making data-flow tracing and attribution harder. Beefed up Topic 3 security content with ten slides woven into the existing sections: multi-layer bias mitigation; the four structural security gaps of agents (multi-step inputs, tool chaining, opaque execution, untrusted entities); direct vs indirect prompt injection; jailbreaking and jailbreak persistence; a new 'Advanced Threats — A Look Ahead' section (data poisoning, backdoor attacks, AI supply-chain risk, agent-to-agent attacks, misalignment / Goodhart's Law); and human-in-the-loop at machine speed. Schedule timing and TSC K/A text unchanged.	Dr Alfred Ang
4.13	26 August 2026	Practical instrument renamed to Case Study across the assessment set, slides, Lesson Plan and Learner Guide. Each Case Study task trimmed to TWO questions: Task 1 keeps the generative-vs-agentic-vs-agent and useful-vs-not-trusted questions; Task 2 keeps the weak-vs-strong prompt and prompt-principles questions; Task 3 keeps the leaky-vs-guarded chatbot and accountability questions. Marks rebalanced (20/24/26, total 70). Schedule timing and TSC K/A text unchanged.	Dr Alfred Ang
4.14	26 August 2026	Learning Outcomes slide now carries the full accredited LO1-LO3 wording. Activity overview slides for Activities 1, 4, 6, 7 and 8 regenerated with activity-specific step flows and 'You'll produce' lines matching the activity packs and Learner Guide. Thank You slide layout corrected. Activity PDFs re-stamped to the current version. Schedule timing and TSC K/A text unchanged.	Dr Alfred Ang
4.15	26 August 2026	Learner Guide and Lesson Plan realigned to the v4.15 deck: Topic 3 job redesign restructured into a narrative sequence (Why Now, The Stakes, Our Operating Model, The Problem, The Solution, The Intervention, The Product, The Worked Example), all slide references renumbered to the 168-slide deck, and Activity 6 extended with the six simulator personas and the Activity 6-Coaching-Sarah worked transcript.	Dr Alfred Ang
4.16	26 August 2026	Lesson Plan slide references aligned to the 168-slide v4.14 deck: the deck total corrected from 165 to 168, and a Slides column added to the one-day schedule giving the exact deck range for every session (all 168 slides covered, no gaps or overlaps).	Dr Alfred Ang
5.0	7 November 2026	Full overhaul. Deck redesigned in the Agentic AI Applications with Claude Code theme. Topic 1 and Lab 1 reuse the Human-AI Workforce with Autonomous AI Agents course slides, with this course's code, plus K2, K3 and A4 slides. Topic 2: prompt injection, jailbreak and data leak. Topic 3: the five principles of	Dr Alfred Ang

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		AI security, governance and incident response. Six labs replace the eight activities, each in its own folder with mock data, scripts and a PDF of prompt instructions, plus a copy-prompt web page. Case Study rewritten as Q1 to Q3 with lab screenshots as answers. Written Assessment questions unchanged; duration aligned to the approved Course Proposal (30 + 30 minutes).	

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Course Information

Item	Detail
Course Title	AI Security for Autonomous AI Agents
WSQ Course Reference	TGS-2025060473
TSC	Generative AI Principles and Applications (ICT-INT-0052-1.1), Level 1
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN: 201200696W)
Duration	1 day, 8 hours: 7 hours of instruction including tea, plus 1 hour of assessment
Daily Timing	9:00 am to 6:00 pm, with a 1-hour lunch that is not counted
Mode	Classroom Facilitated Training. Learners bring a laptop
Course scenario	Kopi Kaki Co., a fictional Singapore kopi startup. All lab data is fictional
Slide deck	WSQ - Master Trainer Slides - TGS-2025060473 - AI Security for Autonomous AI Agents-v50.pptx (131 slides)
Trainer	Dr Alfred Ang

Learning Outcomes

- LO1: Demonstrate generative AI concepts and applications relevant to customer service and hospitality management.
- LO2: Apply prompt engineering techniques and analyse output variations to improve generative AI performance in service settings.
- LO3: Identify ethical risks and analyse bias in AI-generated content used in customer engagement.

Knowledge and Ability Statements

Code	Statement
K1	Importance of data quality, preprocessing, model pipeline and model training (e.g., impact of data bias from training data)
K2	Underlying principles, core concepts and theories governing generative AI
K3	Difference between generative and discriminative models
K4	Impact of prompt engineering on the model outputs of generative AI
K5	Generative AI model workings, including training data, algorithms, and outputs
A1	Analyse limitations and potential biases in AI-generated content
A2	Identify the ethical implications and societal impact of AI-generated content
A3	Apply understanding of generative AI principles to use cases
A4	Demonstrate the use of generation AI in diverse applications (e.g., summarisation, inference, reasoning, transformation of content, augmentation of content)
A5	Analyse generative AI models' performance metrics and evaluate the influence of prompt variations

Learning Units (approved Course Proposal)

LU	Learning Outcome	Course Proposal topics	K & A	Delivered through	Allocated
LU1: Generative AI Fundamentals	LO1: Demonstrate generative AI concepts and applications relevant to customer service and hospitality management.	T1: Underlying principles, core concepts and theories governing generative AI T2: Difference between generative and discriminative models T3: Demonstrate the use of generation AI in diverse	K2, K3, A4	Topic 1: Human-AI Collaboration and Always-On AI Agents	120 min

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		applications (e.g., summarisation, inference, reasoning, transformation of content, augmentation of content)			
LU2: Prompt Engineering	LO2: Apply prompt engineering techniques and analyse output variations to improve generative AI performance in service settings.	T1: Importance of data quality, preprocessing, model pipeline and model training (e.g., impact of data bias from training data) T2: Impact of prompt engineering on the model outputs of generative AI T3: Apply understanding of generative AI principles to use cases T4: Analyse generative AI models' performance metrics and evaluate the influence of prompt variations	K1, K4, A3, A5	Topic 2: AI Agent Vulnerabilities	120 min
LU3: Ethical Considerations	LO3: Identify ethical risks and analyse bias in AI-generated content used in customer engagement.	T1: Generative AI model workings, including training data, algorithms, and outputs T2: Identify the ethical implications and societal impact of AI-generated content T3: Analyse limitations and potential biases in AI-generated content	K5, A1, A2	Topic 3: AI Governance and Security	180 min
Assessment	All	Written Assessment (SAQ) (30 min) and Case Study (30 min)	K1–K5, A1–A5	5:00 pm to 6:00 pm	60 min

Reconciliation to the Course Proposal

LU	Method	CP minutes	Scheduled	Match
LU1	Interactive presentation	60 min	60 min	Yes
LU1	Peer teaching / Peer practice	30 min	30 min	Yes
LU1	Case studies	30 min	30 min	Yes
LU2	Interactive presentation	60 min	60 min	Yes
LU2	Peer teaching / Peer practice	30 min	30 min	Yes
LU2	Case studies	30 min	30 min	Yes
LU3	Interactive presentation	100 min	100 min	Yes
LU3	Peer teaching / Peer practice	40 min	40 min	Yes
LU3	Case studies	40 min	40 min	Yes
Assessment	Written Exam	30 min	30 min	Yes
Assessment	Case Study	30 min	30 min	Yes

Tea breaks sit inside the presentation time of the LU they fall in. Lunch (60 minutes) is not counted.
Total: 420 minutes of instruction plus 60 minutes of assessment.

Course Schedule

Time	Duration	Topic / Activity	LU	Method	Slides
9:00–9:10	10 min	Welcome, digital attendance, trainer, ground rules, Learning Outcomes, LMS and assessment briefing	LU1	Interactive presentation	1–16
9:10–9:50	40 min	Topic 1: From rules to learning to action;	LU1	Interactive	17–28

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		four engineering lenses 2023 to 2026; agentic AI versus AI agents; how an agent works like a person; the agentic loop; how generative AI writes; generative versus discriminative models; AI does the work, humans decide		presentation	
9:50–10:20	30 min	Topic 1: Five Human-AI collaboration scenarios; use cases; AI for science; OpenClaw to Hermes to hosted always-on agents; the Big Five always-on agents compared	LU1	Case studies	29–48
10:20–10:30	10 min	Tea break	LU1	-	-
10:30–11:00	30 min	Lab 1: Your always-on CEO assistant. Screenshot for Case Study Q1	LU1	Peer teaching / Peer practice	49–56
11:00–11:30	30 min	Topic 2: Prompt engineering and data quality; definition and examples of prompt injection; why a model cannot tell a genuine prompt from a malicious one; consequences	LU2	Interactive presentation	57–66
11:30–12:00	30 min	Lab 2: Install Hermes Desktop on a free model, simulate prompt injection, fix it. Screenshots for Case Study Q2	LU2	Case studies	67–74
12:00–12:15	15 min	Topic 2: Definition and examples of jailbreak; techniques; what stronger models and guardrails change	LU2	Interactive presentation	75–79
12:15–12:30	15 min	Lab 3: Simulate a jailbreak, fix it. Screenshots for Case Study Q2	LU2	Peer teaching / Peer practice	80–85
12:30–13:30	60 min	Lunch	-	-	-
13:30–13:45	15 min	Topic 2: Definition and examples of data leak; personal data and the PDPA	LU2	Interactive presentation	86–89
13:45–14:00	15 min	Lab 4: Simulate a data leak, fix it. Screenshots for Case Study Q2	LU2	Peer teaching / Peer practice	90–96
14:00–15:10	70 min	Topic 3: How a model works; limits, bias and ethics; the five principles of AI security explained, with examples	LU3	Interactive presentation	97–112
15:10–15:20	10 min	Tea break	LU3	-	-
15:20–15:30	10 min	Topic 3: The frameworks behind the five principles: IMDA, NIST, OWASP, ISO, PDPA	LU3	Interactive presentation	113
15:30–16:10	40 min	Lab 5: Create an AI Governance Policy from an interview prompt. Screenshot for Case Study Q3	LU3	Peer teaching / Peer practice	114–118
16:10–16:20	10 min	Topic 3: Incident response under the PDPA: Contain, Assess, Report, Evaluate	LU3	Interactive presentation	119
16:20–17:00	40 min	Lab 6: Create an AI Incident Report from mock agent traces. Screenshot for Case Study Q3	LU3	Case studies	120–125
17:00–17:30	30 min	Written Assessment (SAQ): 5 questions, K1 to K5	Asses s	Written Exam	126–131
17:30–18:00	30 min	Case Study: Q1 to Q3, paste your lab screenshots as your answers, A1 to A5	Asses s	Case Study	126–131

Lab Reference

Lab	Title	Allocated	LO, Ability	Evidence for	Slides
Lab 1	Your Always-On CEO Assistant: Meta Muse or OpenAI Dots	30 min	LO1, A4	Case Study Q1	49–56

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Lab 2	Prompt Injection: One Fake Note in a Customer Review	30 min	LO2, A3, A5	Case Study Q2	67–74
Lab 3	Jailbreak: 'It's Only a Comedy Skit'	15 min	LO2, A3, A5	Case Study Q2	80–85
Lab 4	Data Leak: 'Just Send the Agency a Sample'	15 min	LO2, A3, A5	Case Study Q2	90–96
Lab 5	Create an AI Governance Policy: Your Agent Interviews You	40 min	LO3, A1, A2	Case Study Q3	114–118
Lab 6	Create an AI Incident Report: The Leak From Three Months Ago	40 min	LO3, A1, A2	Case Study Q3	120–125

Lab materials: one folder per lab under labs/, holding LAB-0N-Instructions.pdf, data/ and scripts/. Prompts can be copied from prompts/index.html, also published at https://tertiarycourses.github.io/TGS-2025060473-AI-Security-for-Autonomous-AI-Agents/prompts/index.html.

Resources

- Learner laptop (Mac on Apple Silicon, or Windows) with Hermes Desktop
- Free Nous Portal account; a ChatGPT plan for the stronger model where available; Meta Muse account
- Training Google account seeded with the Lab 1 mock meetings and emails, prepared by the trainer
- Slide deck, Learner Guide, lab PDFs and the copy-prompt page

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